

Failure chain (measured)

1. Storage intact

needle content decodable
at its source position
L8 probe acc 0.995–1.000
incl. every failing prompt



2. Transport collapses

16 retrieval heads (of 256)
needle attention on failing
prompts: 0.432 → 0.187
localized: spec. $p < 0.0005$



3. Readout degrades

answer-position decodability
0.714 (wrong) vs 0.960 (right)
accuracy 0.938 → 0.688
(256 → 3,840 tokens)



Symptom

router specialist
starvation
partial $p = 0.64$
(not repairable)

Repair (causal, controlled)

Label-free span detection

IDF-weighted lexical overlap,
question vs context windows
no forward pass · hit rate 100%
on all needle substrates

detected span



Attention boost repair

pre-softmax bias β at the 16 heads
onto the detected span
OLMoE 14/14, Granite 5/5 repaired
99–101% of oracle effect

reopens the collapsed heads

